(How) Will Stablecoins Disrupt Banking?

Bank competition, financial stability and the changing uses of money

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Stablecoins and banking: two propositions

Cong (2025), Deposit Dynamics, Financial Stability, and Regulatory Design

- **1. Bank competition** Stablecoins need not reduce deposits or lending. Bank repricing and the GENIUS framework can limit yield-driven disintermediation.
- **2. Financial stability** The principal risk mechanisms are familiar. Reserve, redemption and operational design determine how they transmit.
- **Scope** Fully reserved, fiat-backed payment stablecoins under a prudential framework.

White paper prepared with support from Coinbase, Paradigm, PayPal and Stripe; conclusions are the author's.

Deposit competition changes banks' behavior

- **Initial friction** Deposit-market power and customer switching costs allow banks to pay less than the competitive return.
- **Bank response** A credible outside option can induce higher deposit rates and better services.

- **Intermediation** Under the competition mechanism, deposits and lending can expand as franchise margins narrow.
- **Model provenance** The white paper adapts the CBDC benchmark of Chiu et al. (2023).

The competition benchmark has three regions

Qualitative mechanism in the white paper's Figure 2

Outside-option remuneration	Banking response in the benchmark
Low	Little change in deposit pricing or intermediation
Moderate	Bank repricing can raise deposits and lending
High	Direct substitution can dominate the competition effect

Competition can expand deposits and lending before substitution dominates.

GENIUS constrains yield-driven deposit competition

Provision / mechanism	Implication for the white paper's argument
Issuer interest/yield restriction	Limits direct interest competition with bank deposits
At least 1:1 eligible liquid reserves	Constrains reserve risk and reserve-funded returns
Redemption and disclosure framework	Supports confidence in the payment claim
Remaining adoption margins	Convenience, distribution and rewards need separate analysis

GENIUS narrows direct yield competition while leaving convenience and distribution as adoption margins.

Recalibrating bank competition to the post-2020 rate environment

2024 observed rates

Return on reserves	**5.21%**
Average deposit rate	**1.66%**
Difference	**3.55 pp**

Bank deposit rates rose much less than returns on reserves.

Updated inputs in Cong (2025)

Production curvature	**0.697**
Reserve requirement	**0%**
Return on reserves (average)	**3.9%**
Inflation / money growth	**4.2%**
Cash / digital / mixed (%)	**3.27 / 31.9 / 64.83**

Maintained structure: 26 banks, trading probability 0.56, and the remaining baseline parameters.

Competition can expand banking intermediation

Illustrative 2025 calibration, conditional on the maintained model and institutional regime

Remuneration region	Deposit and lending response reported in the paper
Low	Approximately unchanged
About 4–6%	Competition supports deposits and lending
Above about 6%	Substitution can dominate and reduce intermediation

The 4–6% region supports intermediation under the paper's stated calibration.

The principal financial risk mechanisms are familiar

Stablecoin concern	Familiar financial mechanism
Redemption run or depeg	Runs and coordination problems in redeemable claims
Reserve or custodian impairment	Asset, counterparty and custody exposure
Redemption faster than asset realization	Liquidity and settlement mismatch
Concentrated reserve sales	Market-capacity limits and fire-sale spillovers
Failure to transfer or reconcile claims	Operational risk in payment infrastructure

Established prudential principles provide a starting point for design.

Common shocks can propagate through onchain links

- **Timing** Continuous token transfers can meet constraints in fiat redemption and settlement.
- **Concentration** Large flows may depend on shared custodians, dealers and operating providers.
- **Scale** Greater normal-time participation can coexist with larger common stress exposures.
- **Research question** How do connected issuers, banks, protocols and markets amplify a common shock?

The research task is to trace shocks through connected issuers, banks and protocols.

Prudent integration can improve banking and payments

Potential gain	Condition for realizing it
Better payment services	Reliable access, redemption and interoperability
Less settlement exposure	Enforceable completion and sufficient liquidity
Bank service opportunities	Fees cover custody, operating and liquidity costs
More efficient intermediation	Competition improves terms without unmanaged risk

Payment innovation and banking can reinforce one another when the underlying exposures are managed.

Five Facts About Stablecoins

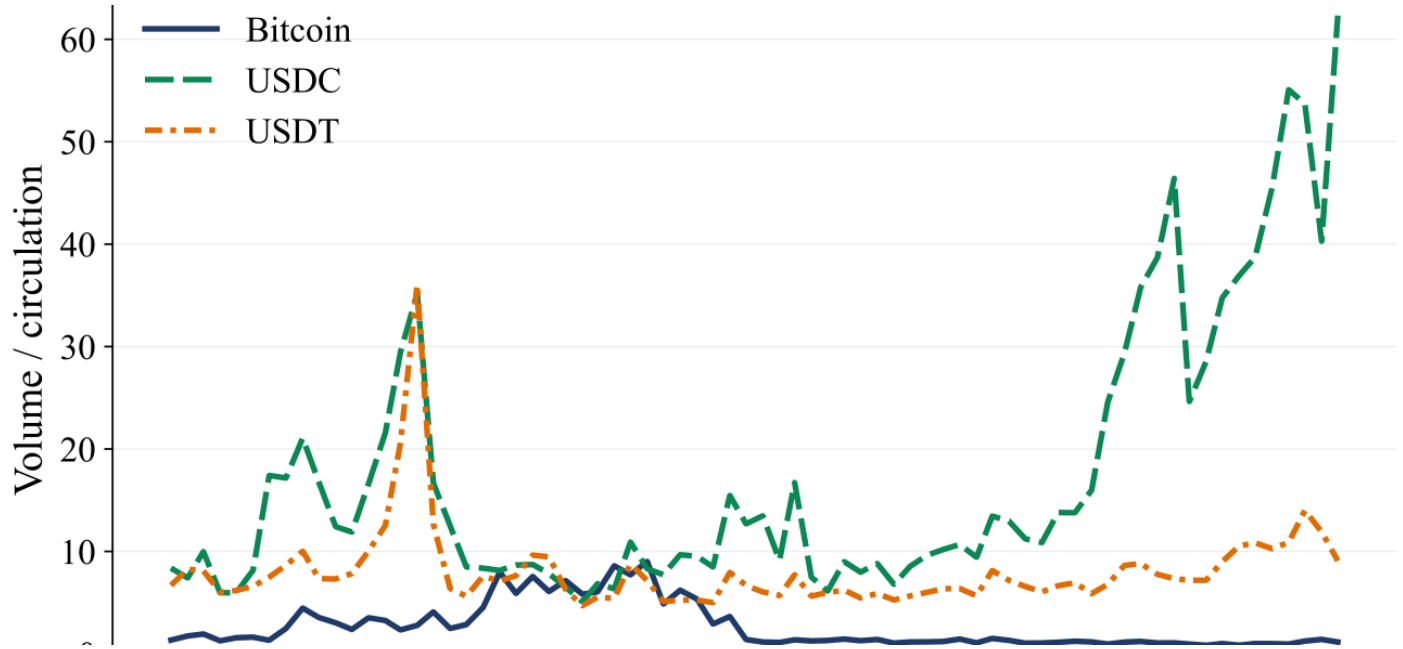
Cong, Liao, Prasad & Zhang (2026) · Wallet-level evidence from 2020–2025

Fact	Data / method	Banking margin
1. Transaction currency	Transfers and size distributions	Payment services
2. Global adoption	Timing-based location probabilities	Cross-border funding
3. Cash management	Wallet moments and a structural benchmark	Demand for balances
4. Onchain lending	Protocol and platform borrowing	Credit infrastructure
5. Programmable payments	Contract and facilitator classifications	Automated settlement

Cong et al. (2026), Sections 3–4 and Appendices A–D. Gordon Liao is affiliated with Circle

Fact 1 — Stablecoins increasingly function as money

Paper Figure 4 · Monthly onchain volume relative to outstanding circulation



62

USDC monthly onchain velocity, Dec. 2025

9

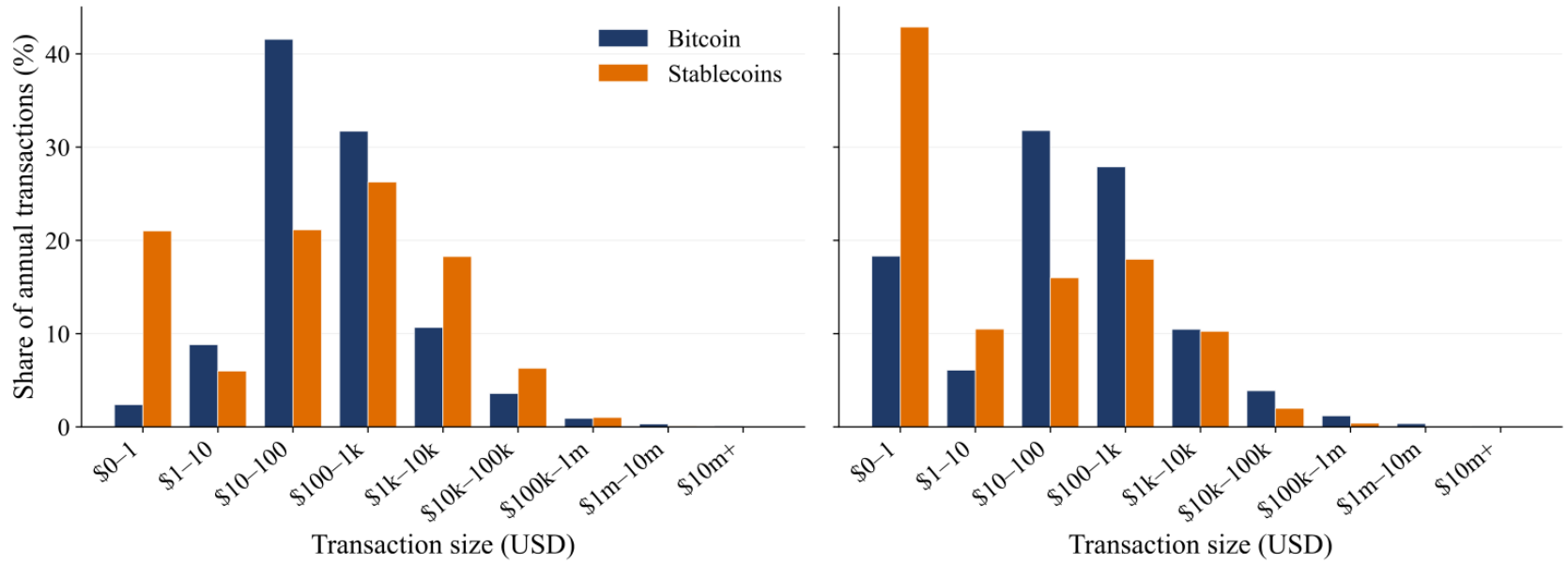
USDT monthly onchain velocity, Dec. 2025

Gross onchain turnover has grown rapidly, particularly for USDC.

Cong, Liao, Prasad & Zhang (2026), Figure 4. Source: supplied Five Facts About Stablecoins draft.

Fact 1 — Small stablecoin transfers have multiplied

Paper Figure 5 · Transaction-size distribution, filtered USDC/USDT and Bitcoin



21% → 43%

Share of transfers below $1, 2022 to 2025

53%

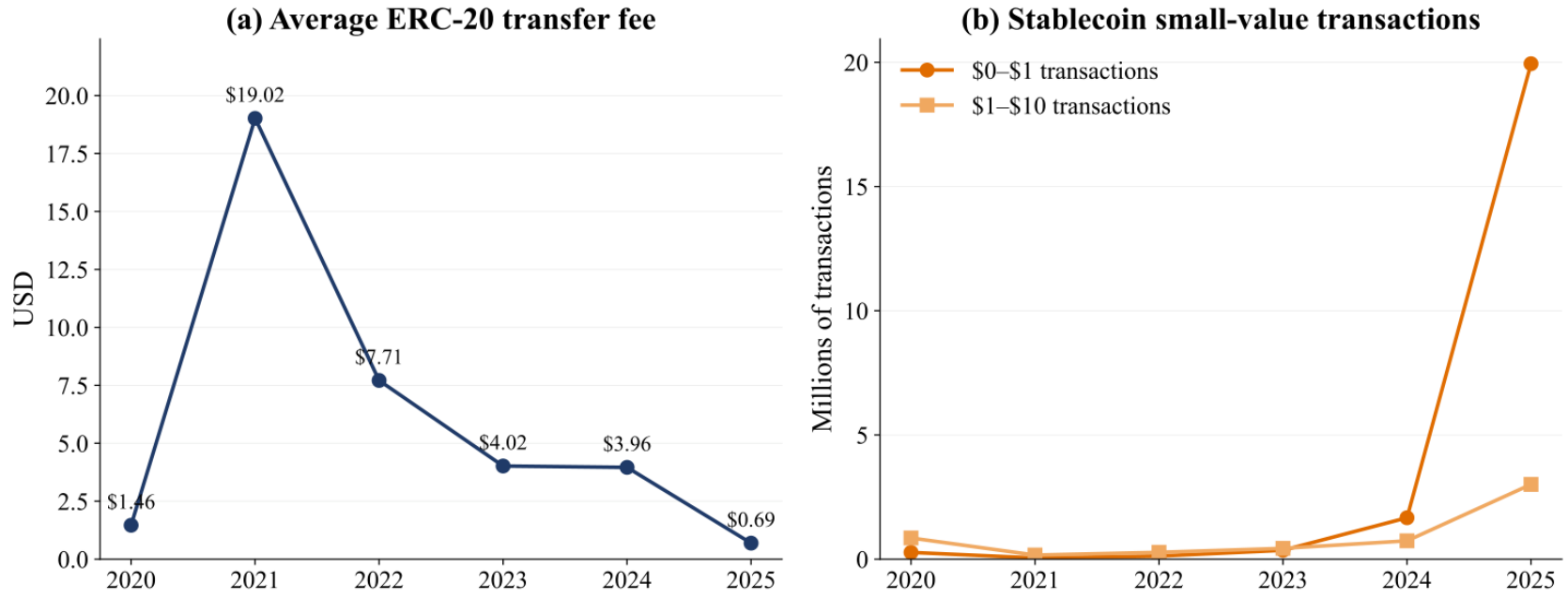
Share below $10 in 2025

The transaction distribution has shifted toward retail and micropayment amounts.

Cong, Liao, Prasad & Zhang (2026), Figure 5. Source: supplied Five Facts About Stablecoins draft.

Fact 1 — Lower fees support micropayments

Paper Figure 6 · Ethereum transfer cost and small-value stablecoin transfers



$19.02 → $0.69

Average Ethereum ERC-20 fee, 2021 to 2025

19.9m

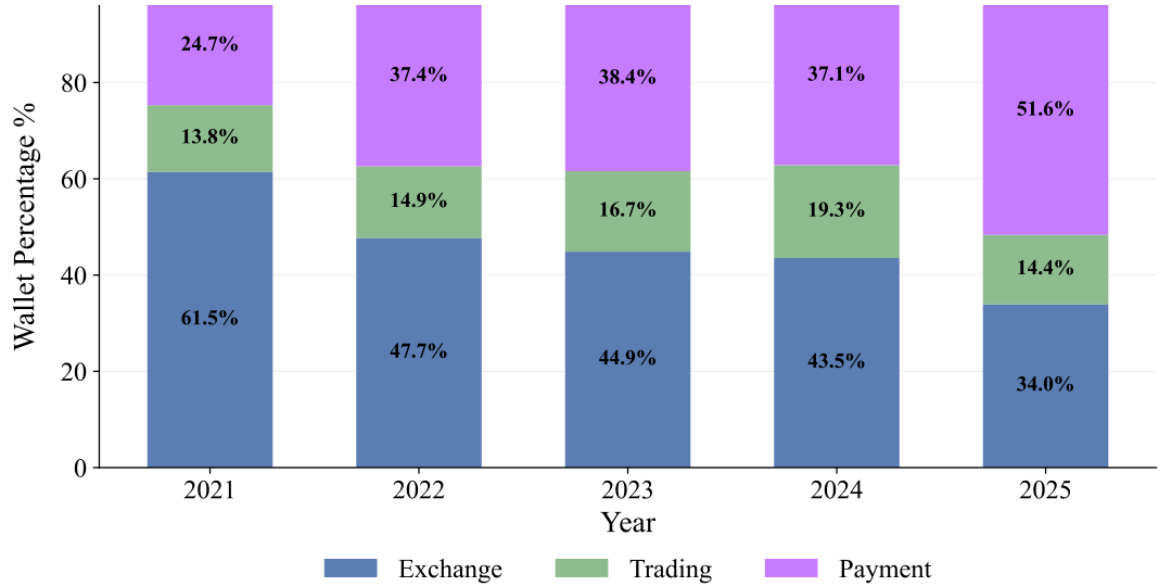
Ethereum stablecoin transfers below $1 in 2025

Falling network costs expand the set of viable payment sizes.

Cong, Liao, Prasad & Zhang (2026), Figure 6. Source: supplied Five Facts About Stablecoins draft.

Fact 1 — Wallet use is shifting toward payments

Paper Figure 8 · Behavioral classification of active Ethereum wallets



25% → 52%

Payment-type wallets, 2021 to 2025

61% → 34%

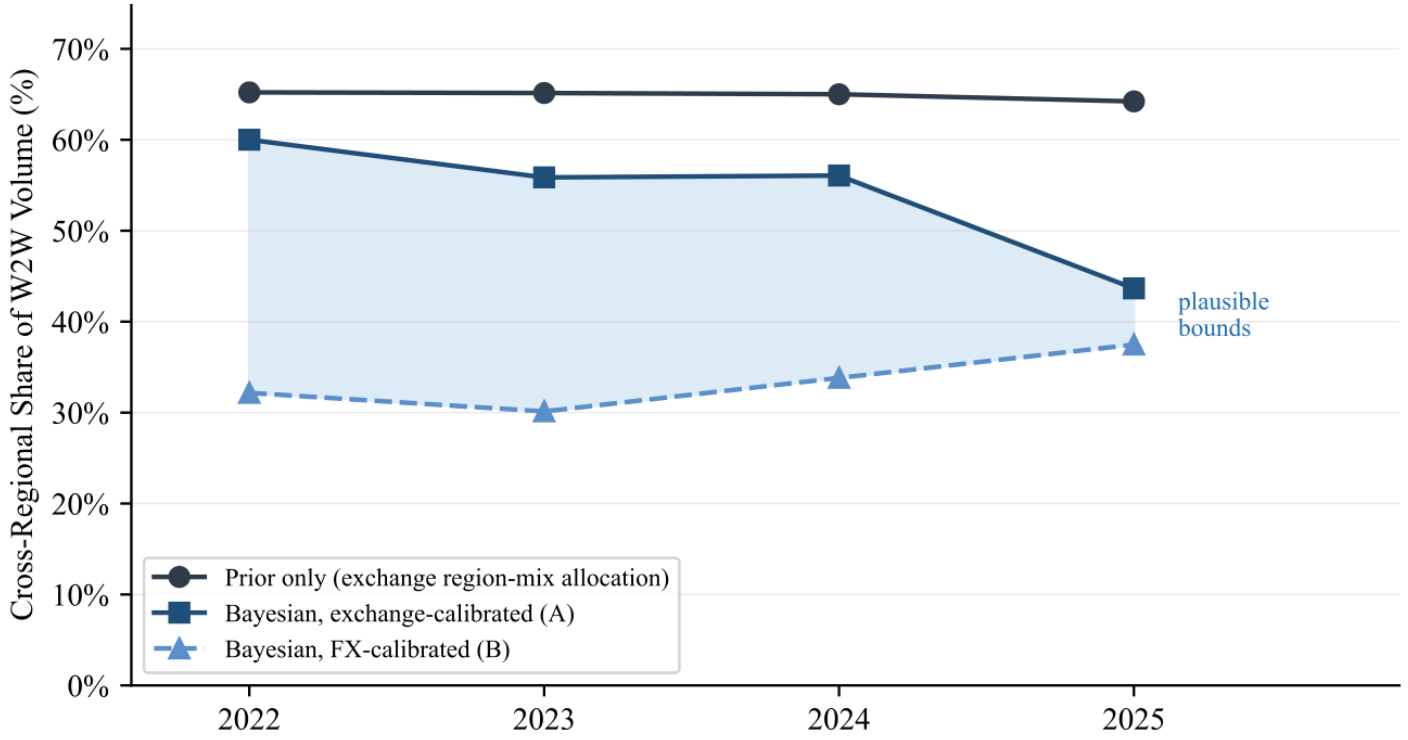
Exchange-type wallets over the same period

More active wallets now transact primarily in stablecoins.

Cong, Liao, Prasad & Zhang (2026), Figure 8. Source: supplied Five Facts About Stablecoins draft.

Fact 2 — Stablecoins are used within and across regions

Paper Figure 10 · Exchange-based prior versus timing-adjusted geographic inference



37–44%

Across-region share in 2025

≈65%

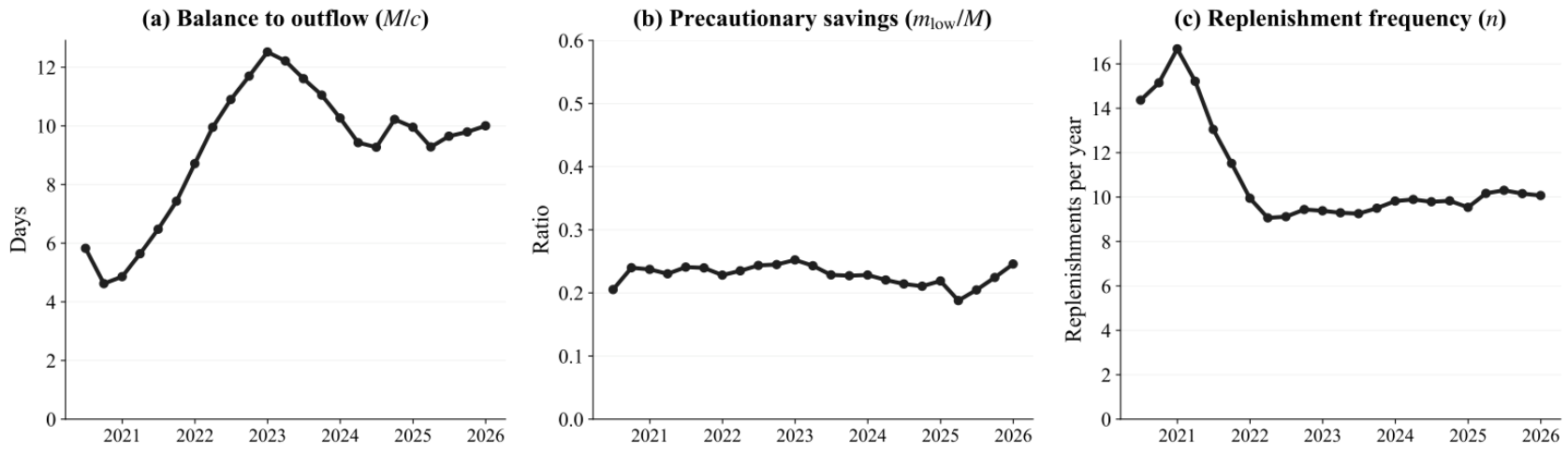
Across-region share using exchange data alone

Transaction timing reveals a larger within-region role than conventional attribution suggests.

Cong, Liao, Prasad & Zhang (2026), Figure 10. Source: supplied Five Facts About Stablecoins draft.

Fact 3 — Wallet balances resemble cash inventories

Paper Figure 12 · Observed money-demand moments from Ethereum wallet balances



10 days

Typical balance / daily outflow, late 2025

≈10/year

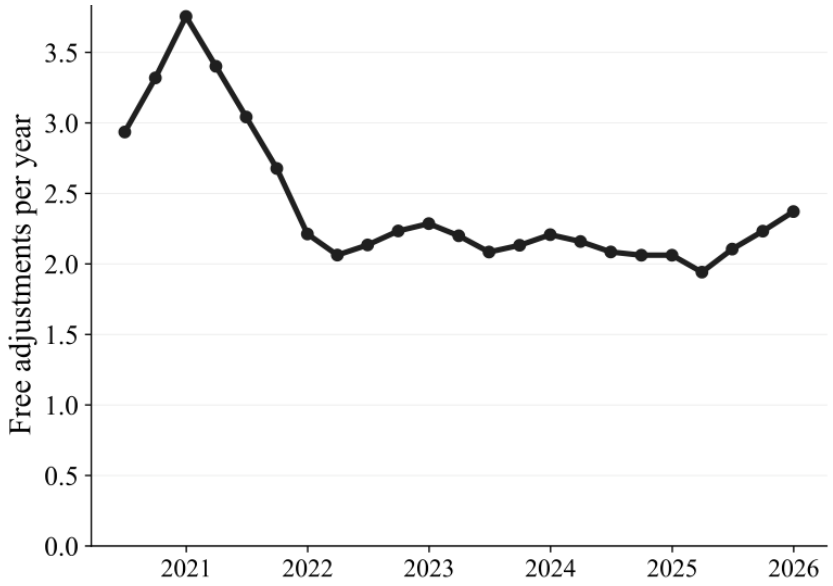
Replenishments by a typical wallet

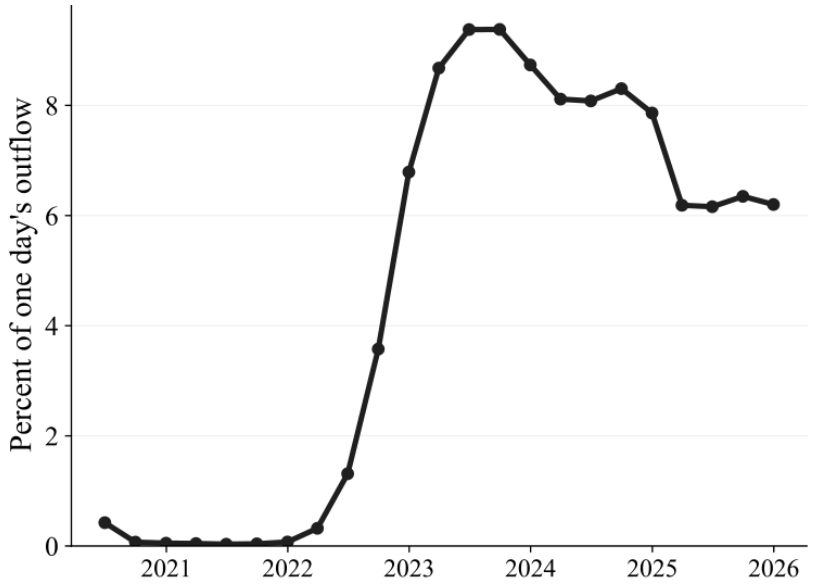
Users carry working balances and replenish them periodically.

Cong, Liao, Prasad & Zhang (2026), Figure 12. Source: supplied Five Facts About Stablecoins draft.

Fact 3 — Stablecoin wallet access has become cheaper

Paper Figure 13 · Alvarez–Lippi structural cash-management parameter estimates





2.4/year

Estimated free top-up opportunities in late 2025

9.4% → 6.2%

Estimated fixed top-up cost, peak 2023 to late 2025

The model points to improving access, especially among transaction-oriented wallets.

Cong, Liao, Prasad & Zhang (2026), Figure 13. Source: supplied Five Facts About Stablecoins draft.

Fact 4 — Onchain credit markets are growing

Paper Figure 14 · Secured protocol and selected unsecured onchain credit



4.1×

Secured loans: $9.3bn to $37.8bn, end-2023 to end-2025

5.7×

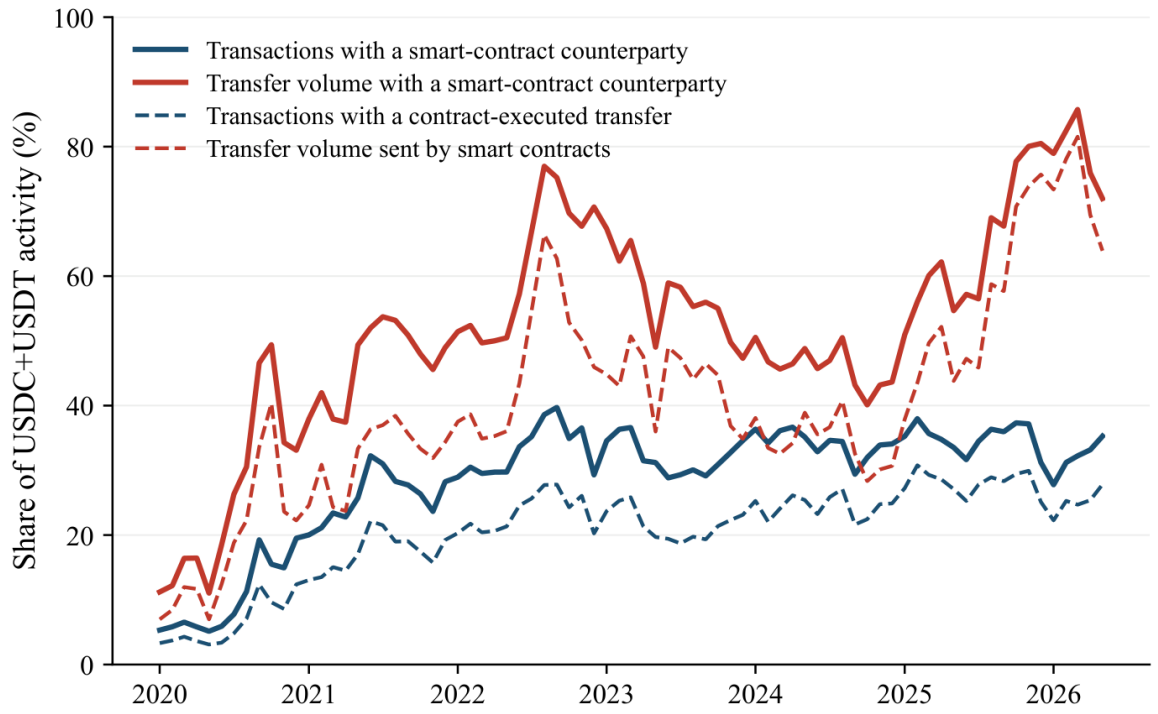
Selected institutional loans: $0.38bn to $2.15bn, Apr–Dec 2025

Onchain credit brings programmable lending, collateral and liquidation rules.

Cong, Liao, Prasad & Zhang (2026), Figure 14. Source: supplied Five Facts About Stablecoins draft.

Fact 5 — Smart contracts move most stablecoin value

Paper Figure 15 · Contract participation in Ethereum USDC/USDT transfers



35%

Ethereum transfer transactions involving contracts, 2025

69%

Share of gross transfer value involving contracts

Most current programmable settlement is financial automation.

Cong, Liao, Prasad & Zhang (2026), Figure 15. Source: supplied Five Facts About Stablecoins draft.

Fact 5 — Machine micropayments are feasible

Paper Figure 16 · x402 monthly facilitator payments, May 2025–May 2026



68m

x402 transactions at the Dec. 2025 peak

7.6m

Transactions in May 2026 after a sharp retrenchment

x402 demonstrates micropayments; sustained commercial adoption remains to be established.

Cong, Liao, Prasad & Zhang (2026), Figure 16. Source: supplied Five Facts About Stablecoins draft.

Five facts, and their implications for banking

Observed margin	Banking object still to identify
Payment activity	Useful purchases, all-in costs and customer relationships
Wallet geography	Displaced asset and full funding chain
Balance management	Stable funding versus transitory settlement balances
Onchain credit	Borrower substitution, maturity, collateral and losses
Programmable transfers	Delegation, durable use and contingent liquidity

Next research questions: network-wide stress transmission and the quality of delegated agent decisions.

Banks’ deposit rates influence stablecoin demand

Ahmed & Aldasoro (2026) · U.S. weekly data, January 2019–June 2025

Change in deposit rate

+1 bp

An increase in the rate paid by banks

Stablecoin growth over ten weeks

–2.8 pp

Aggregate OLS estimate

USDC –4.8 pp

USDT –1.9 pp

Local projections controlling for market rates, Bitcoin, dollar and VIX; N = 324 at 10 weeks.

Deposit repricing influences stablecoin demand; the credit response depends on banks’ equilibrium adjustments.

Ahmed & Aldasoro (2026), BIS Working Paper 1382, Table OA.A.1 and §5.

Payment information, liquidity and monetary trust

Contribution	Economic margin	Question for banking
Parlour, Rajan & Zhu	Payment information	Which projects obtain finance?
Lee & Tou	Issuer-related liquidity demand	How does incoming funding change bank portfolios?
Harris & Rogoff	Institutions behind private money	What sustains trusted par redemption?

The mechanisms connect bank funding, information in payment flows and trust in private money.

e-CNY in 2026: remuneration with bank-deposit treatment

Commercial-bank wallet arrangements effective January 1, 2026

Design dimension	Documented arrangement
Liability	Commercial-bank deposit treatment
Remuneration	Posted demand-deposit rate for eligible identified wallets
Prudential treatment	Normal reserves and deposit-insurance framework
Inference for banking	Adoption and bank participation can change jointly

Interest rises from zero. The sources do not establish a yield premium over ordinary deposits.

PBOC policy reported by gov.cn, Dec. 29, 2025. CCB announcement, Dec. 31, 2025; Type IV excluded

Agentic payments and banks' liquidity services

- **Cash management** Automated adjustment may reduce idle payment balances and increase replenishment frequency.
- **Liquidity timing** Common execution rules may concentrate payments and redemptions at the same time.
- **Banking services** Credit lines, cash management and settlement access can remain valuable even when payment execution moves onchain.
- **Evaluation question** Which delegated agent policies remain robust to liquidity shocks and strategic responses?

Delegated payments: credit and enforceable claims

Agent or tokenized-asset need	Intermediation problem
Spending authority and budget	Identify the principal and enforce limits
Contingent credit	Underwrite repayment and fund drawdowns
Collateral and settlement	Verify ownership, valuation and legal priority
Failure or dispute	Allocate losses and provide enforceable recourse

Delegation links payment authority to credit, ownership and enforceable claims.

Cong, Mayer & Rabetti (2026), Tokenizing Real-World Assets. AP2 is an architecture example

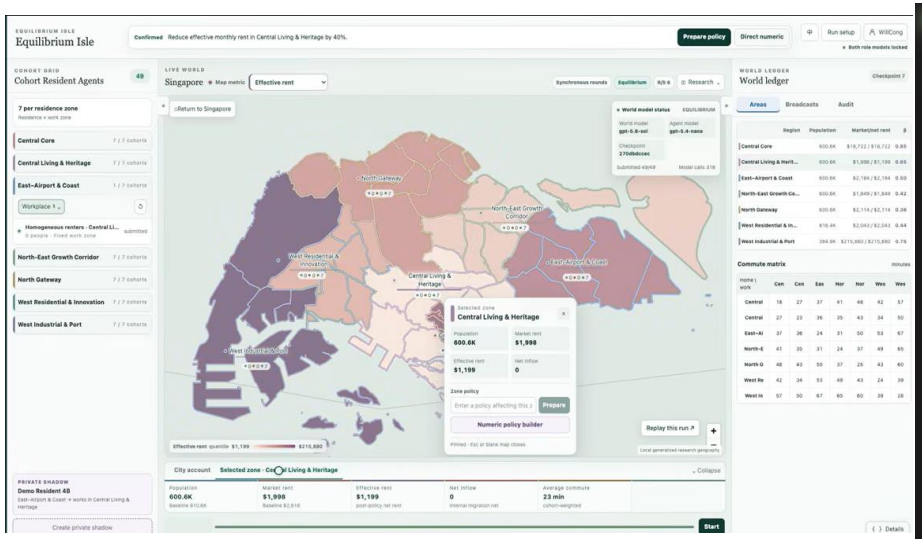
Banking in an onchain, agentic economy

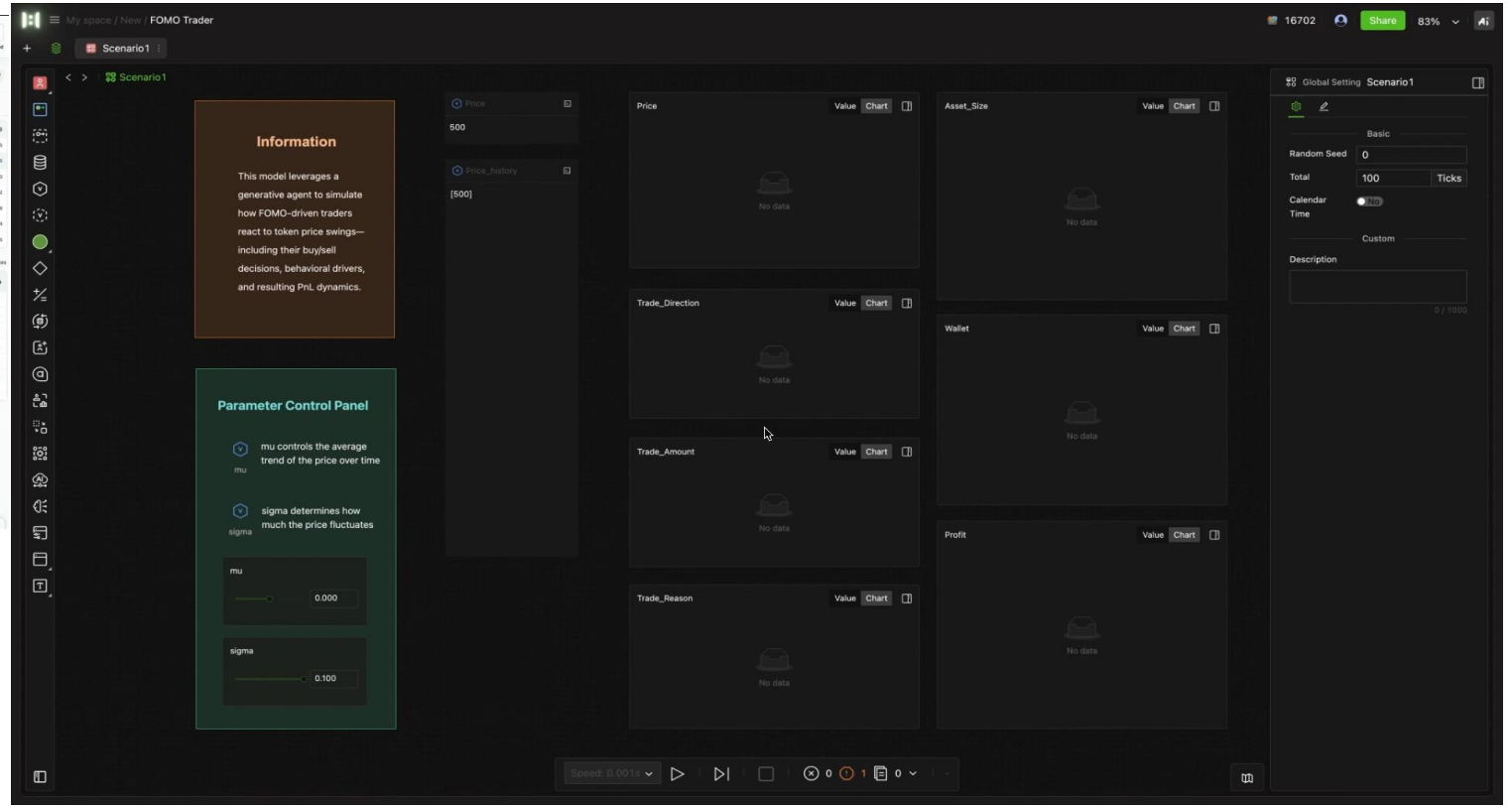
- **Bank competition** Deposit pricing, reserve design and market structure determine how stablecoins affect bank credit.
- **Financial stability** Runs, liquidation and settlement strains propagate across connected institutions.
- **Economic World Models** Model systemic interactions and test safeguards under plausible stress scenarios.
- **Global MetaCase** Benchmark and improve autonomous agents acting under economic and institutional constraints.

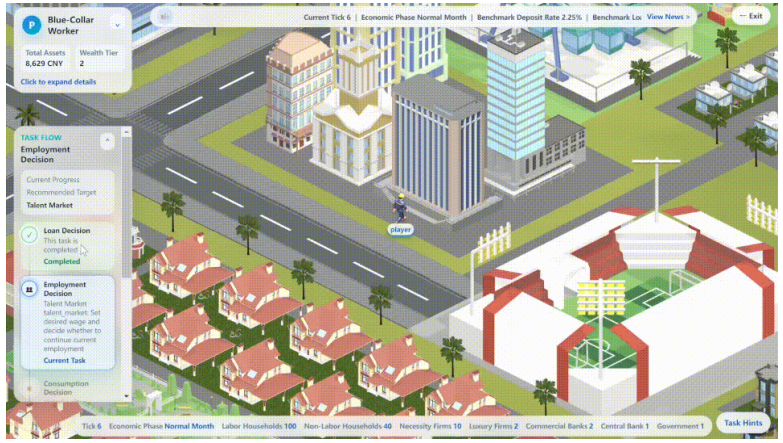
Cong (2025); Cong, Liao, Prasad & Zhang (2026). EWM and MetaCase are proposed research directions.

Economic World Models Initiative

Economic World Models (EWM) | an integrated laboratory for mechanisms and stress propagation







Global Metacase Initiatives

Global MetaCase Initiative | economic decision quality, incentives and institutional constraints

MetaCase: state and data → objective and authority → decision → outcome

Global Metacase Challenge

Scientific test: does individual competence survive interaction with other autonomous agents?

